



Liver & Extrahepatic Biliary Apparatus

Histology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 10 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

Which is the largest gland in the body?

- A. Gallbladder
- B. Sublingual gland
- C. Parotid gland
- D. Liver
- E. Pancreas

2.

EASY

The connective tissue capsule of the liver is called:

- A. Perisinusoidal space
- B. Lamina propria
- C. Portal canal
- D. Muscularis mucosae
- E. Glisson's capsule

3.

EASY

Which structures lie at the corners of a classic liver lobule?

- A. Kupffer cells
- B. Hepatic veins
- C. Portal triads
- D. Central veins
- E. Sinusoids

4.

EASY

Which cell is the primary structural and functional unit of the liver?

- A. Hepatocyte
- B. Kupffer cell
- C. Ito cell
- D. Endothelial cell
- E. Bile duct cell

5.

MEDIUM

CASE

In a classic lobule, plates of hepatocytes radiate outward from a thin-walled vessel in the centre of the hexagon.

Which vessel is this?

- A. Portal vein branch
- B. Lymphatic vessel
- C. Bile ductule
- D. Central vein
- E. Hepatic arteriole

6.

MEDIUM

The space of Disse lies between:

- A. the capsule and the peritoneal mesothelium
- B. a bile canaliculus and a bile duct
- C. a portal triad and the central vein
- D. two adjacent hepatocytes side by side
- E. the hepatocyte and the sinusoidal endothelium

7.

MEDIUM

CASE

A cell on the luminal surface of a sinusoid has an ovoid nucleus, is derived from monocytes and contains engulfed material.

Which cell is this?

- A. Endothelial cell
- B. Kupffer cell
- C. Hepatocyte
- D. Ito cell
- E. Bile duct cell

8.

MEDIUM

CASE

A cell in the space of Disse contains lipid droplets and has long contractile processes that can regulate sinusoidal blood flow.

Which cell is this?

- A. Ito (stellate) cell
- B. Monocyte-derived cell
- C. Hepatocyte cell
- D. Kupffer cell
- E. Endothelial cell

9.

MEDIUM

Which hepatocyte surface exchanges materials with the blood?

- A. The sinusoidal surface
- B. The capsular surface
- C. The intercellular surface
- D. The perisinusoidal endothelium
- E. The canalicular surface

10.

MEDIUM

Bile canaliculi are limited by:

- A. a ring of Kupffer cells
- B. the membranes of two hepatocytes
- C. a fenestrated endothelium
- D. a layer of cuboidal cells
- E. a simple columnar epithelium

11.

MEDIUM

CASE

The liver parenchyma is divided into interlocking triangles, each with a portal triad at its centre and a central vein at each of its three corners.

Which model is this?

- A. The portal lobule
- B. The central lobule
- C. The hepatic acinus
- D. The classic lobule
- E. A liver lobe

12.

MEDIUM

What fraction of hepatic blood does the portal vein supply, and how would you describe it?

- A. ~50%; oxygen-rich, nutrient-rich
- B. ~20%; oxygen-rich, nutrient-poor
- C. ~80%; oxygen-poor, nutrient-rich
- D. ~20%; oxygen-poor, nutrient-rich
- E. ~80%; oxygen-rich, nutrient-poor

13.

MEDIUM

Which statement about bile acids is correct?

- A. About 90% are made in hepatocyte smooth ER; they digest protein
- B. All are made in the gallbladder; they absorb water
- C. About 10% are made in hepatocyte smooth ER; they emulsify lipids
- D. About 10% come from the intestine; they store vitamin A
- E. None are made in the liver; they phagocytose bacteria

14.

MEDIUM

Bile ductules (canals of Hering) are best described as:

- A. fenestrated sinusoids of the acinus
- B. grooves limited by two hepatocytes
- C. thick-walled ducts with a sphincter
- D. columnar tubes containing goblet cells
- E. cuboidal channels joining canaliculi to bile ducts

15.

HARD

CASE

A researcher traces a droplet of bile from the hepatocyte outward to the liver hilum.

Which order is correct?

- A. Canaliculus → bile duct → ductule → common hepatic duct → hepatic duct
- B. Hepatic duct → bile duct → ductule → canaliculus → common hepatic duct
- C. Canaliculus → ductule → bile duct → hepatic duct → common hepatic duct
- D. Ductule → canaliculus → bile duct → common hepatic duct → hepatic duct
- E. Bile duct → canaliculus → ductule → hepatic duct → common hepatic duct

16.

HARD

CASE

In a hepatic acinus, three zones are marked from the line joining the portal triads to the central veins.

Which hepatocytes are exposed to the highest oxygen, nutrient and toxin levels, and why?

- A. Zone II only, because it lies at the acinus centre
- B. Zone III, because the hepatic artery ends at the central vein
- C. All zones equally, because sinusoids are fenestrated
- D. Zone III, nearest the central vein, where blood arrives first
- E. Zone I, nearest the portal triads, where blood arrives first

17.

HARD

CASE

After chronic liver injury the space of Disse fills with collagen and other extracellular matrix.

Which cell is responsible, and what is its normal storage role?

- A. Ito cells; storing iron in ferritin
- B. Kupffer cells; storing iron in ferritin
- C. Hepatocytes; storing vitamin A in lipid droplets
- D. Ito (stellate) cells; storing vitamin A in lipid droplets
- E. Endothelial cells; storing collagen

18.

HARD

CASE

A thrombus completely blocks the portal vein.

What fraction of the normal liver blood supply is lost, and which organs' venous blood no longer reaches the liver?

- A. ~50%; blood from the intestine only
- B. ~20%; blood returning from the hepatic veins
- C. ~80%; blood from the kidney and heart
- D. ~80%; blood from the pancreas, spleen and intestine
- E. ~20%; blood from the pancreas, spleen and intestine

19.

HARD

CASE

The lower end of the bile duct joins the main pancreatic duct, and the shared channel opens into the duodenum.

Which name and regulator apply to this common channel?

- A. Cystic duct, regulated by the sphincter of Oddi
- B. Ampulla of Vater, regulated by the sphincter of Oddi
- C. Hilum, regulated by CCK receptors
- D. Common hepatic duct, regulated by Kupffer cells
- E. Ampulla of Vater, regulated by the muscularis mucosae

20.

HARD

CASE

A pear-shaped organ has a highly folded mucosa of simple columnar cells with microvilli and no goblet cells, two or three irregular smooth-muscle layers, and a serosa (an adventitia where it touches the liver).

Which organ is this, and what is its function?

- A. Appendix; houses lymphoid nodules
- B. Gallbladder; stores and concentrates bile
- C. Duodenum; absorbs nutrients
- D. Stomach; secretes pepsinogen
- E. Common bile duct; secretes bile acids

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	The liver is the largest gland of the body.
2	E	The liver is covered by a CT capsule (Glisson's), thicker at the hilum.
3	C	Portal triads sit at each corner; the central vein is in the middle.
4	A	Hepatocytes are the primary structural and functional unit of the liver.
5	D	Hepatocyte plates radiate outward from the central vein in the middle of the lobule.
6	E	The narrow perisinusoidal space of Disse separates the liver cell surface from the endothelial lining; hepatocyte microvilli extend into it.
7	B	Monocyte-derived phagocytes on the luminal side of the sinusoidal endothelium are Kupffer cells.
8	A	Lipid-droplet (vitamin A) fat-storing cells of the space of Disse are Ito / stellate cells.
9	A	Sinusoidal-surface microvilli project into the space of Disse and exchange materials with the blood.
10	B	A canaliculus is formed by grooves in the membranes of neighbouring hepatocytes — no separate epithelium.
11	A	The portal lobule follows bile flow: a triangle with a portal triad at its centre and a central vein at each corner.
12	C	Portal vein $\approx$ 80% of hepatic blood (from pancreas, spleen, intestine); hepatic artery $\approx$ 20% with higher oxygen.
13	C	About 90% derive from intestinal absorption, $\sim$ 10% are synthesized in the SER; bile acids emulsify lipids for lipase and absorption.
14	E	Canaliculi drain into cuboidal ductules (canals of Hering), which drain into the bile ducts of the portal triad.
15	C	Bile flows canaliculi $\rightarrow$ ductules (Hering) $\rightarrow$ bile ducts in the portal triad $\rightarrow$ right/left hepatic ducts $\rightarrow$ common hepatic duct.
16	E	Zone I lies closest to the terminal distributing vessels; Zone III, nearest the central vein, sees blood after it has been used.
17	D	Ito cells, located in the space of Disse, are a major collagen source during injury and store vitamin A in lipid droplets.
18	D	The portal vein carries $\sim$ 80% of hepatic blood, conveying blood from the pancreas, spleen and intestine.
19	B	The lower bile duct joins the main pancreatic duct as the common hepatopancreatic duct (ampulla of Vater), regulated by the sphincter of Oddi.
20	B	Highly folded simple columnar mucosa with microvilli, no goblet cells, 2–3 irregular smooth-muscle layers = gallbladder, which stores and concentrates bile.

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 4, Dr Hero K. Mustafa
Written from the lecture slides only – check any answer you doubt against your lecture notes.